Order-Agnostic Autoregressive Diffusion Models: A Cross-Modal Study on Images and Categorical Tabular Data

Jorge Barcenilla S. Prieto
Probabilistic Machine Learning, Universidad Carlos III de Madrid

***Abstract*—Order-Agnostic Autoregressive Diffusion Models (OA-ARDMs) unify order-agnostic autoregressive modelling with absorbing discrete diffusion through a single masked-prediction objective. We implemented this objective and applied it, without altering its core, to two distinct modalities: binarized MNIST images and the fully categorical UCI Mushroom dataset. The tabular case required only changes to the input parametrization—per-column vocabularies enforced through masked logits and a dedicated absorbing token—leaving the training and sampling algorithms identical. For each modality we trained two network families and measured held-out negative log-likelihood in bits per dimension (bpd) together with sample fidelity. Convolutional and multilayer-perceptron backbones reached 0.139 and 0.609 bpd on the MNIST and Mushroom test sets, produced recognizable digit samples, and reproduced categorical marginals with a mean total-variation distance below 0.1. The results show that a single order-agnostic objective generalizes across heterogeneous data types with only modality-specific input handling.**

I. Introduction

Deep autoregressive models factorize a joint distribution over D variables into a product of conditionals $p_\theta(x) = \prod_d p_\theta(x_d \mid x_{<d})$, achieving exact likelihoods but imposing a single fixed generation order and requiring D sequential network evaluations to sample [2], [5]. Order-agnostic autoregressive models relax the fixed order by treating it as a latent permutation [3], while discrete diffusion models destroy data toward an absorbing state and learn the reverse process [4]. The Order-Agnostic Autoregressive Diffusion Model (OA-ARDM) unifies these views: a single shared network is trained, in the spirit of BERT, to predict an arbitrary masked subset of variables, yielding flexible order and adaptive parallel sampling [1].

However, the formulation and its public demonstrations are framed almost exclusively around homogeneous modalities—images and text—in which every variable shares one vocabulary and a regular spatial or sequential structure. Whether the same objective transfers cleanly to heterogeneous categorical tabular data, where each column has a different cardinality and no spatial adjacency, is not addressed by the original parametrization.

In this paper, we present an implementation of the OA-ARDM and apply its unmodified training and sampling algorithms to two modalities: binarized MNIST and the categorical UCI Mushroom dataset. We compare two network families per modality and evaluate them under a common likelihood metric (bits per dimension) and under sample-fidelity criteria, isolating the few changes the tabular setting demands.

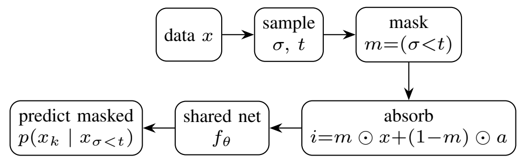

Fig. 1. One training/sampling step of the OA-ARDM. The same objective and network are reused across both modalities; only the input parametrization (a, the vocabulary, and f_θ) is modality-specific.

II. Method

A. Order-agnostic diffusion objective

Let $x \in \{0, \ldots, K-1\}^D$ and let σ be a permutation of $\{0, \ldots, D-1\}$ drawn uniformly. For a timestep $t \sim \mathcal{U}(1, \ldots, D)$ the binary mask $m = (\sigma < t)$ marks the variables already revealed. Masked variables are replaced by an absorbing value a, giving the network input $i = m \odot x + (1-m) \odot a$. A single shared network f_θ predicts the categorical parameters of every masked variable, and training maximizes the one-term estimator of the likelihood bound

$$\log p(x) \;\geq\; \mathbb{E}_{t\sim\mathcal{U}(1,\ldots,D)}\big[\, D \cdot L_t \,\big], \tag{1}$$

where L_t is the average log-likelihood over the masked variables at step t [1]. Sampling reverses the process: starting from a fully absorbed vector, variables are revealed one at a time along σ, each drawn from the network's predicted categorical (Fig. 1).

B. Two instantiations

The objective above is modality-agnostic. For images, x is a binarized 28×28 digit (D=784, K=2); the absorbing value is 0 and f_θ is either a convolutional network with a time embedding (LeNetWithTime) or a small vision transformer over 7×7 patches (TinyTimeViT). For tabular data, x is a row of D=22 categorical columns of differing cardinality (2–12), which breaks the single-vocabulary assumption. We use one shared vocabulary of size $K = \max_j K_j + 1$ whose last index is a dedicated absorbing token (avoiding collision with a valid category), and we enforce per-column validity by masking the logits of categories absent from a column to $-\infty$. Each column is embedded as a token (value, column-identity,

TABLE I
HELD-OUT NEGATIVE LOG-LIKELIHOOD (BITS PER DIMENSION) AND TABULAR SAMPLE FIDELITY.

Data	Model	Train	Val	Test	TVD
MNIST	LeNetWithTime	0.139	0.140	**0.139**	—
	TinyTimeViT	0.164	0.160	0.159	—
Mushroom	MLPWithTime	0.603	0.613	**0.609**	0.17
	TabTransformer	0.677	0.683	0.667	**0.075**

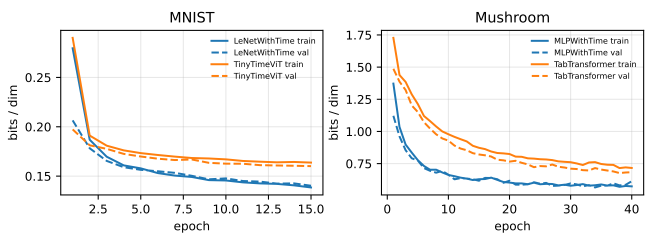

Fig. 2. Training (solid) and validation (dashed) bits per dimension. Left: MNIST. Right: Mushroom.

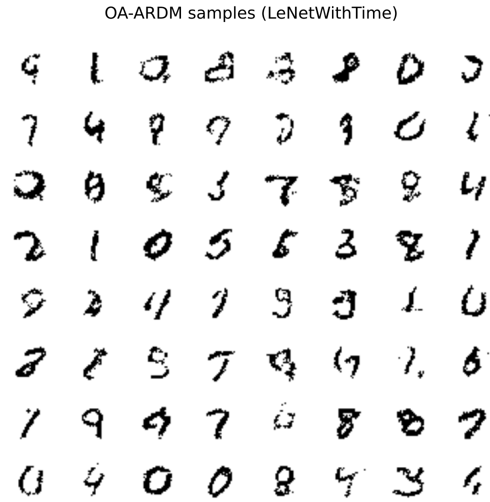

Fig. 3. Unconditional samples generated by the order-agnostic sampler (LeNetWithTime, binarized MNIST).

and mask embeddings) and processed by either a multilayer perceptron (MLPWithTime) or a Transformer encoder over the column tokens (TabTransformer). Training and sampling are otherwise identical to the image case.

III. Experimental Setup

MNIST was binarized at threshold 0.5 and split into 55,000 training, 5,000 validation and the standard 10,000 test images. Mushroom was encoded column-wise (missing values as their own category, the constant `veil- type` column dropped) with the edible/poisonous label modelled as an additional column, and split 70/15/15 stratified on the label. All models were trained with Adam (learning rate 10^{-3}): 15 epochs for MNIST, 40 for Mushroom. We report negative log-likelihood as bits per dimension, $\mathrm{bpd} = \mathrm{NLL}_{\mathrm{nats}}/(D \ln 2)$, estimating the stochastic objective with K Monte-Carlo repetitions per held-out batch. Sample fidelity for the tabular models is summarized by the mean per-column total-variation distance (TVD) between 2,000 generated rows and the test set.

IV. Results

Table I reports train, validation and test bits per dimension for all four models, together with the tabular sample-fidelity metric; in every case train and test values differ by less than 0.01. Per-label tabular test bpd for the MLP was 0.629 (edible) and 0.588 (poisonous). Figure 2 shows the optimization trajectories, Figure 3 the unconditional MNIST samples, and Figure 5 the generated versus empirical category marginals for both tabular models.

V. Discussion

The objective was met: an unaltered order-agnostic diffusion core produced calibrated likelihoods and plausible samples in both an image and a categorical tabular setting, with the tabular adaptation confined to the input parametrization. The MNIST test bpd of 0.139 is consistent with reported order-agnostic results on binarized MNIST [1], supporting the correctness of the implementation. In both modalities the smaller backbone (convolutional or perceptron) attained lower likelihood than its transformer counterpart, which we attribute to the modest model size and training budget relative to the transformer's capacity. Likelihood and sample fidelity did not coincide on the tabular data: the MLP achieved the best bpd yet the TabTransformer reproduced categorical marginals more faithfully (Fig. 5), a known decoupling between log-likelihood and perceptual or distributional sample quality.

These findings are bounded by the small networks, the single training seed, and a sample-fidelity metric that captures only first-order (marginal) structure; higher-order dependencies between columns were not quantified. Even within these bounds, the central observation holds across two unrelated datasets: a single order-agnostic diffusion objective transfers between images and heterogeneous categorical tables, requiring only a per-column vocabulary and a dedicated absorbing token rather than any change to training or sampling.

VI. Conclusion

We implemented an Order-Agnostic Autoregressive Diffusion Model and showed that its training and sampling algorithms apply unchanged to two distinct modalities, binarized images and fully categorical tabular data, once the input parametrization accounts for per-column vocabularies through masked logits and a dedicated absorbing token. Across both datasets the model produced competitive held-out likelihoods and faithful samples, demonstrating that a single order-agnostic objective is a genuinely modality-portable foundation for discrete generative modelling.

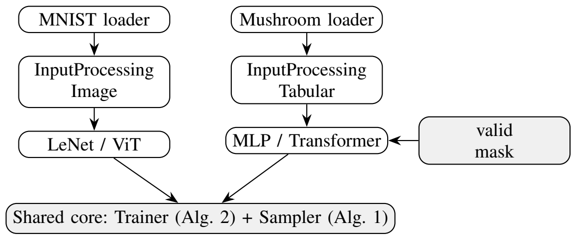

Fig. 4. Module layout: parallel image and tabular front-ends share one training/sampling core.

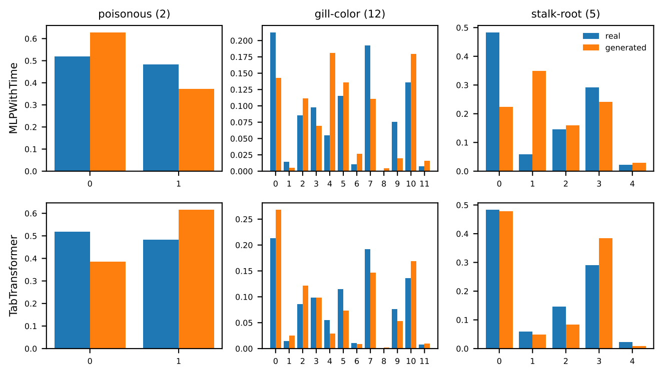

Fig. 5. Empirical (blue) versus generated (orange) category marginals on the Mushroom test set for three representative columns. Top: MLPWithTime. Bottom: TabTransformer.

Appendix A
Software Architecture

The implementation isolates a shared algorithmic core from modality-specific front-ends (Fig. 4). Two parallel branches—image and tabular—each provide a data loader, an input-processing module that builds the masked, time-conditioned features, and a backbone network. Both branches feed the same two algorithms: the training objective (Algorithm 2) and the order-agnostic sampler (Algorithm 1). The tabular branch adds a precomputed per-column validity mask used to project logits onto the categories that each column admits. The pipelines are reproducible end-to-end (`main_mnist.py`, `main_tabular.py`) and the tabular core is covered by unit tests for masking, the absorbing token, loss back-propagation and sampler validity.

References

[1] E. Hoogeboom *et al.*, "Autoregressive Diffusion Models," *ICLR*, 2022.
[2] M. Germain *et al.*, "MADE: Masked Autoencoder for Distribution Estimation," *ICML*, 2015.
[3] B. Uria *et al.*, "A Deep and Tractable Density Estimator," *ICML*, 2014.
[4] J. Austin *et al.*, "Structured Denoising Diffusion Models in Discrete State-Spaces," *NeurIPS*, 2021.
[5] A. van den Oord *et al.*, "Pixel Recurrent Neural Networks," *ICML*, 2016.
[6] J. Schlimmer, "Mushroom Data Set," UCI Machine Learning Repository, 1987.